
RansKnow: An Open Dataset of Ransomware-Related YouTube Transcripts with MITRE ATT&CK Annotations

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Abstract

1 **[TODO]** Draft, not final. Should state: (1) what RansKnow is (1,128 video records,
2 1,034 transcripts, 92 cybersecurity YouTube channels, MITRE ATT&CK-aligned
3 regex features); (2) the paper’s actual contribution is arguably the label-quality audit
4 methodology as much as the corpus itself – Section 4 found that a naive keyword-
5 matched family label collided with ordinary English words (play, hive, cactus,
6 ...) badly enough to inflate the single most-detected ransomware family by $\sim 3.5\times$,
7 and Section 9’s counterfactual analysis shows a trained classifier partially inherited
8 that same word-level fragility even after the fix; (3) state plainly, in the abstract
9 itself, not just in a limitations section: baseline results (Section 5) are weak-label
10 agreement scores, not independently verified accuracy – this project does not
11 include human-annotated ground truth (see the scope note in Section 4.3); (4)
12 headline OOD-robustness numbers, which don’t depend on that caveat.

13 1 Introduction

14 **[TODO]** The gap: no open, transcript-level ransomware knowledge corpus with MITRE ATT&CK-
15 aligned annotations. Motivate why YouTube security content specifically (volume, diversity of source
16 – vendor threat-intel channels, DFIR conferences, independent researchers – versus e.g. only vendor
17 blog posts). State the paper’s contributions as a list: (1) the corpus; (2) the label-quality audit
18 methodology and its concrete findings (Section 4); (3) the experiment framework and weak-label
19 baseline results (Section 5), reported as agreement with the audited extraction pipeline – state directly
20 here, not just in Limitations, that no independent human-annotated ground truth was collected for
21 this project (Section 4.3); (4) out-of-distribution robustness results (Section 8), which are not subject
22 to that caveat; (5) counterfactual and perturbation-robustness interpretability results (Section 9).

23 2 Related Work

24 **[TODO]** Three threads: (a) existing ransomware/threat-intel datasets (structured IOC feeds, MITRE
25 ATT&CK mapping efforts, prior ransomware incident corpora – note what’s specific to RansKnow:
26 raw natural-language transcripts rather than structured feeds); (b) weak/distant supervision in security
27 NLP (keyword/regex-based labeling, its known failure modes); (c) conformal prediction and OOD
28 detection in NLP, as background for Sections 7 and 8 once written.

29 3 Dataset Construction

30 RansKnow [Kabuye, 2026] consists of 1,128 video records sourced from 92 cybersecurity YouTube
31 channels, of which 1,034 have an extracted transcript with real content; the remaining 94¹ could not
32 be transcribed by any method in the fetch cascade.

33 3.1 Channel selection

34 Channels were curated in two batches. Batch 1 (50 channels, C01–C50) covers established DFIR
35 vendors, security conferences, and independent researchers. Batch 2 (51 candidate channels, C51–
36 C101) expanded coverage into threat-intelligence vendors, additional conferences, and OT/ICS-
37 security and government channels; 42 of the 51 candidates yielded usable transcripts, 1 was a
38 confirmed duplicate of an existing Batch 1 channel, 4 had no discoverable dedicated YouTube
39 channel, and 4 resolved to real channels with zero ransomware-keyword-matching content in their
40 upload history.

41 **[TODO]** Cite the specific channel registry files (rubrics/Dataset_Channel_Registry_*.xlsx)
42 and consider a full channel table as a supplementary table rather than inline.

43 3.2 Video selection

44 Within each channel, videos were selected by scanning recent uploads for a fixed keyword list
45 (ransomware, extortion, lockbit, incident response, ...; full list in Appendix A) matched
46 against title and description, capped at 20 videos per channel and a minimum duration of 5 minutes.

47 **[TODO]** Report the actual keyword list and discuss selection bias explicitly: this is a convenience
48 sample conditioned on channel operators' own titling and description-writing behavior, not a random
49 sample of ransomware discourse.

50 3.3 Transcript acquisition

51 Transcripts were acquired via a three-stage cascade: (1) the YouTube Data API's official captions
52 where available; (2) yt-dlp's auto-generated captions; (3) local Whisper transcription as a fallback.
53 **This cascade matters more than a footnote:** only 74 of 1,034 transcripts (7.2%) came from official
54 captions. 959 (92.7%) are ASR output – 554 from whisper_base (the smallest, least accurate
55 Whisper checkpoint) and 405 from a local Whisper fallback, 1 from yt-dlp captions.

56 **[TODO]** This is Finding B from the label-quality audit (Section 4). Decide whether to present it
57 once, here, or duplicate the framing in both sections – probably present the raw fact here and the
58 downstream implication (jargon mistranscription risk) in Section 4.

59 4 Label-Quality Audit

60 The Knowledge Agent pipeline (Scripts/knowledge_agent.py) extracts structured features from
61 every transcript via regex matching: ransomware family mentions (against a maintained alias list),
62 MITRE ATT&CK tactic mention counts across 10 tactics, offensive tool mentions across 8 tools, and
63 platform signals (Windows/Linux/ESXi). Three concrete problems were found and are described
64 here rather than left as an implicit caveat, because a downstream classifier trained on these labels
65 inherits whatever bias they contain.

1

[TODO] Verify this figure against the live dataset before submission – $1128 - 1034 = 94$, but confirm this is failed transcript extraction and not something else, e.g. videos excluded post-fetch.

66 4.1 Finding A: generic-word alias collisions

67 Nine of 41 family aliases collide with common English or business words (play, hive, cactus,
68 agenda, clop, inc, maze, medusa, royal) – identified programmatically via dictionary lookup
69 against /usr/share/dict/words, then confirmed by manual transcript inspection (e.g. hive
70 matching “registry hive,” a routine DFIR term, rather than Hive ransomware). A windowed-context
71 disambiguation fix (requiring a ransomware-context term within ~ 15 tokens of the match) was applied
72 and reduces false-positive volume substantially without eliminating it – residual false positives still
73 occur when context vocabulary is itself dense (see Scripts/audit_family_aliases.py for the
74 audit methodology and Table 1 for before/after counts).

Table 1: Before/after family mention counts for the 9 flagged aliases, windowed-context disambiguation applied. Play alone accounted for 386 of 494 (78%) of all family-tagged videos before the fix.

Family	Before	After	Dropped
Play	386	111	275
Qilin	88	21	67
Hive	23	10	13
Maze	11	5	6
Royal	28	19	9
INC	10	7	3
Cactus	9	6	3
Cl0p	11	11	0
Medusa	7	7	0

75 4.2 Finding B: ASR-dominated transcript quality

76 As noted in Section 3.3, 92.7% of transcripts are ASR output rather than official captions.

77 **[TODO]** Run the Phase 0.4 spot-check (manual comparison of ASR output against source audio for
78 a fixed jargon term list – LockBit, Mimikatz, PsExec, Cobalt Strike, rclone, BloodHound –
79 across the whisper-transcribed subset) and report a measured word-error-rate-on-jargon figure here.
80 Aggregate detection-rate comparisons across Transcript_Provider did not show an obvious gap,
81 which is not the same as no effect – individual mistranscriptions of rare jargon can matter without
82 moving an aggregate rate.

83 4.3 Scope: no independent ground truth

84 An earlier design stage of this project planned a held-out, independently annotated evaluation subset
85 (180 videos, stratified by channel type, year, and current family-count) specifically to break the
86 circularity of evaluating a classifier against the same weak-label pipeline it was trained on. That
87 subset was never annotated. External annotator recruitment required institutional ethics approval
88 not available to this project; a single-investigator pass was considered as a fallback and rejected as
89 still too close to the system under evaluation to serve as independent ground truth in a way worth
90 claiming in a paper.

91 The consequence is stated here plainly, and repeated at the point every affected result is reported,
92 rather than left as a single Limitations bullet a reader could miss: **every quantitative result in**
93 **Section 5 onward, with the explicit exception of Section 8’s out-of-distribution results, measures**
94 **agreement with the audited Knowledge Agent extraction pipeline (Section 4), not verified**
95 **accuracy against human judgment.** Section 4 demonstrates concretely that this pipeline’s labels
96 contain real, non-trivial error even after the disambiguation fix (Finding A); nothing downstream
97 corrects for that. Readers should treat macro-F1 and calibration numbers in this paper as characterizing
98 how well models reproduce a specific, imperfect, documented extraction process – not as claims
99 about ransomware-domain understanding in an absolute sense. Section 10 returns to this as the
100 paper’s central limitation.

5 Task Definitions and Baselines

Five prediction tasks are defined over the corpus: ransomware family (multi-label, 18 classes after excluding 3 singleton-occurrence classes, see Table 2), dominant MITRE tactic (multi-class, 11 classes), platform signal (multi-class, 4 classes), offensive tool mentions (multi-label, 7 classes after excluding MegaNZ, which has zero occurrences in the corpus), and a ransomware-relevance binary screen (derived, not a stored column: `Family_Count > 0 OR Tactic_Impact > 0`).

Table 2: Task taxonomy.

Task	Type	Classes	Label source
Family	multi-label	18	alias regex match (audited, §4.1)
Dominant tactic	multi-class (+none)	11	max of 10 tactic-pattern counts
Platform signal	multi-class (+none)	4	max of 3 platform-pattern counts
Tool mentions	multi-label	7	per-tool regex match
Ransomware-relevant	binary	2	derived

5.1 Model ladder and validation protocol

Each task is evaluated over three feature representations (structured Knowledge Agent columns, TF-IDF, mean-pooled sentence embeddings²) and five models (the rule-based Knowledge Agent itself as baseline 0, Logistic Regression, Linear SVM, Random Forest, LightGBM), under three cross-validation protocols: stratified random k -fold, GroupKFold by Channel_ID (to detect channel-style leakage – a model can learn “this is CrowdStrike’s narration style” instead of the content), and a temporal split (train pre-2024, test 2024+, since 79% of the corpus is from 2024 onward).

5.2 Results

Per Section 4.3, every number in this section is weak-label agreement – how well a model reproduces the Knowledge Agent’s own extraction, not verified accuracy. The rule-based Knowledge Agent baseline scores a tautological macro-F1 of 1.000 on every task (checked against its own labels by construction; included as baseline 0, not as evidence of anything) and is omitted from Table 3 accordingly.

Table 3: Best stratified-CV macro-F1 per task, across every representation (structured / TF-IDF / embedding) and trained model (LogReg, Linear SVM, Random Forest, GBM) tried. Weak-label agreement, not verified accuracy (§4.3).

Task	Best representation	Best model	Macro-F1
Ransomware-relevant	TF-IDF	GBM	0.903
Dominant tactic	structured	GBM	0.813
Platform signal	TF-IDF	GBM	0.606
Tool mentions	TF-IDF	LogReg	0.200
Family	embedding	LogReg	0.129

Two patterns worth naming, both consistent with Section 9’s independent checks: (1) tree models (GBM in particular) are the strongest model wherever tested, and structured features remain the strongest representation specifically for `dominant_tactic` – unsurprising, since those features are numerically close to the counts the label itself was derived from; (2) `family` and `tool` are weak under every representation and model combination tried (`family` never exceeds 0.13 macro-F1; `tool` is additionally unstable across CV splits, ranging 0.03–0.20). This is not obviously a framework or tuning limitation – Section 9’s length and channel-identity confound checks rule out two of the more common alternative explanations for weak tabular/text-classification performance, leaving weak-label

²384-dim all-MiniLM-L6-v2, chunked at ~200 words per chunk and mean-pooled, computed via a dedicated Python environment (torch 2.5.1) since the reference implementation’s default environment ships an older torch that cannot satisfy `sentence_transformers`’ requirements; see `Scripts/precompute_embeddings.py`.

ceiling (§4.3) and genuine class sparsity (Section 5’s 18-class family / 7-class tool taxonomies, several classes with single-digit support) as the remaining candidate explanations.

[TODO] Full task×representation×model×split table as supplementary material, or as a figure once uncommented below; cite outputs/experiments/phase1_*.csv directly.

6 Uncertainty Quantification

Scope note (§4.3 applies here directly): calibration is normally reported against ground truth – does an 80%-confidence prediction turn out correct roughly 80% of the time. Without independent labels, the only calibration target available here is the weak-label distribution itself, which changes what a well-calibrated result would mean: a model perfectly calibrated against its own training signal has learned to be a confident, well-calibrated echo of a regex, not necessarily a well-calibrated model of ransomware content. This section is retained, with that meaning made explicit at every reported number, because reliability diagrams and Expected Calibration Error against the weak labels are still informative about a narrower question worth answering: does a model’s confidence track its *agreement with the extraction pipeline*, or is it uniformly overconfident regardless of agreement. The latter would be a red flag on its own.

6.1 Calibration against weak labels

For a multiclass task, let $\hat{p}(\cdot | x)$ be the trained model’s predicted class distribution for input x , and let $\hat{c}(x) = \arg \max_c \hat{p}(c | x)$ with confidence $\hat{p}_{\max}(x) = \max_c \hat{p}(c | x)$. Agreement with the weak label $y^{\text{KA}}(x)$ (the Knowledge Agent’s own extraction) is $a(x) = \mathbb{I}[\hat{c}(x) = y^{\text{KA}}(x)]$. For the two multi-label tasks (family, tool), whose models are independent per-class binary classifiers (Section 5), each (sample, class) pair (x, k) is treated as its own calibration point, with confidence $\hat{p}_{\max}(x, k) = \max(\hat{p}_k(x), 1 - \hat{p}_k(x))$ and agreement against the binary weak label thresholded at 0.5.

Reliability is summarized with $B=10$ equal-width confidence bins $\{I_b\}_{b=1}^B$ and the standard Expected Calibration Error,

$$\text{ECE} = \sum_{b=1}^B \frac{|S_b|}{N} |\text{acc}(S_b) - \text{conf}(S_b)|, \quad S_b = \{x : \hat{p}_{\max}(x) \in I_b\}, \quad (1)$$

where $\text{acc}(S_b) = \frac{1}{|S_b|} \sum_{x \in S_b} a(x)$ and $\text{conf}(S_b) = \frac{1}{|S_b|} \sum_{x \in S_b} \hat{p}_{\max}(x)$. Per the scope note above, $a(x)$ measures agreement with y^{KA} , not correctness, so Eq. (1) is a *weak-label ECE*. The headline number below the scope note is the confident-disagreement rate at threshold $\tau = 0.9$,

$$\text{CDR}_{\tau} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}[\hat{p}_{\max}(x_i) > \tau] \cdot (1 - a(x_i)), \quad (2)$$

the closest available analogue to a “confidently wrong” rate without independently verified labels to actually be wrong against.

Each task is evaluated on the 5-fold stratified out-of-fold predictions of its best (feature, model) combination from Table 3, except family, whose best combination (embedding representation) needs the dedicated environment described in Section 5’s footnote and is reported without a calibration curve here.

Figure 1 plots the reliability curves underlying Table 4. Two things worth naming honestly rather than smoothing over: first, per-bin counts are small (single digits to low tens per bin at $N=1034$), so individual bin heights are noisy – the curves are directionally informative, not point-estimate-precise. Second, family’s row has no ECE/CDR: its calibration point is unavailable in this environment (as above), so only the RF tree-vote entropy proxy, computed on the TF-IDF fallback representation, is reported for it, and even that proxy has no “disagree” entries to average – at the per-(sample, class) level, most of family’s 18 classes are negative for most samples, so the majority-class prediction (“absent”) agrees with the weak label on essentially every RF-on-TF-IDF calibration point despite the task’s low macro-F1 in Table 3, which weights all 18 classes equally rather than by prevalence. Both readings are correct simultaneously; they are just answering different questions (per-pair agreement vs. per-class balanced performance).

Table 4: Weak-label calibration (Equations (1) and (2)) and the RF tree-vote entropy proxy (Equation (3)), best (feature, model) per task. N is the number of (sample) or (sample, class) calibration points. Entropy AUROC is how well entropy alone separates agreement from disagreement with the weak label (Algorithm 1).

Task	N	ECE	CDR _{0.9}	Entropy (agree/disagree)	Entropy AUROC
Relevance	1034	0.079	0.071	0.910 / 0.980	0.769
Dominant tactic	1034	0.051	0.043	0.275 / 0.551	0.860
Platform	1034	0.110	0.104	0.717 / 0.769	0.675
Tool (per-label)	7238	0.016	0.016	0.121 / 0.271	0.977
Family (per-label)	1034	–	–	0.095 / –	–

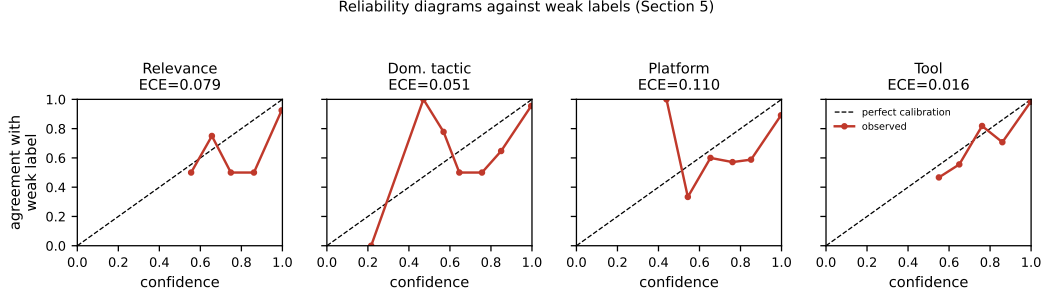


Figure 1: Reliability diagrams against weak labels, one panel per task with a usable calibration curve (Table 4). Diagonal is perfect calibration.

6.2 A free uncertainty proxy: Random Forest tree-vote entropy

Reliability diagrams need a held-out calibration protocol; a Random Forest offers a per-prediction uncertainty signal with no separate calibration step. For an input x and a forest of T trees producing individual (hard) predictions $t_1(x), \dots, t_T(x)$ over K classes, let $\pi_c(x) = \frac{1}{T} \sum_{j=1}^T \mathbb{1}[t_j(x) = c]$ be the empirical vote share for class c . The normalized tree-vote entropy is

$$H(x) = -\frac{1}{\log K} \sum_{c=1}^K \pi_c(x) \log \pi_c(x) \in [0, 1], \quad (3)$$

with $H(x)=0$ when every tree agrees and $H(x)=1$ at a uniform split across all K classes. For the multi-label tasks, $K=2$ per class and $H(x, k)$ is averaged over classes k for a per-sample summary. Algorithm 1 is the exact out-of-fold procedure used to produce Table 4’s entropy columns.

Algorithm 1 RF tree-vote entropy, 5-fold out-of-fold

Require: dataset $\mathcal{D}$, task with label function $y(\cdot)$, feature builder ϕ

- 1: $\mathcal{F} \leftarrow 5$ stratified folds of $\mathcal{D}$
 - 2: **for** fold $(\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{test}}) \in \mathcal{F}$ **do**
 - 3: fit ϕ on $\mathcal{D}_{\text{train}}$; fit Random Forest $\{t_j\}_{j=1}^T$ on $\phi(\mathcal{D}_{\text{train}})$
 - 4: **for** $x \in \mathcal{D}_{\text{test}}$ **do**
 - 5: $t_j(x) \leftarrow$ individual tree prediction, $j = 1, \dots, T$ $\triangleright$ sklearn returns internal integer-encoded classes here, not the original label strings – see note below
 - 6: recover original class labels via the fitted forest’s class index before comparing to $y(x)$
 - 7: compute $H(x)$ via Equation (3); record $a(x) = \mathbb{1}[\text{forest majority vote} = y(x)]$
 - 8: **end for**
 - 9: **end for**
 - 10: **return** $\{(H(x), a(x))\}_{x \in \mathcal{D}}$, pooled across folds
-

The in-line comment in Algorithm 1 records a real implementation pitfall, not a hypothetical one: scikit-learn fits each tree in a `RandomForestClassifier` on internally label-encoded integer tar-

183 gets, so `tree.predict()` returns class *indices*, not the original label strings the ensemble’s own
 184 `.predict()` returns. Comparing tree-level votes directly against string labels without mapping back
 185 through the fitted forest’s class index silently produces 0% apparent agreement on every multiclass
 186 task – caught in this project’s own Phase 2 run (initial numbers showed exactly that pattern across
 187 all three multiclass tasks simultaneously, which is what made it legible as a bug rather than a real
 188 result) and is recorded here as a concrete instance of the caution Section 10 urges about verifying
 189 rather than assuming standard library internals. Table 4’s entropy AUROC column shows the proxy
 190 works well once fixed: 0.68–0.98 across tasks, best on `tool` (many easy true negatives) and weakest
 191 on `platform`.

192 7 Conformal Prediction

193 **Scope note:** split conformal prediction’s distribution-free coverage guarantee is a statement about
 194 exchangeability with the calibration set, not about label correctness – it holds formally even when
 195 calibrated against weak labels, because the guarantee is “the true *weak* label falls in the predicted set
 196 $(1 - \alpha)$ of the time,” not “the true *real-world* label does.” That distinction is the entire content of this
 197 scope note: conformal coverage against the Knowledge Agent’s labels is a well-defined, honestly
 198 reportable number, but it validates agreement with the extraction pipeline, not correctness. Reported
 199 as such.

200 7.1 Method: split conformal with the LAC score

201 Split conformal prediction (Vovk et al., 2005, Lei et al., 2018; see Angelopoulos and Bates, 2023 for
 202 a tutorial treatment) uses a nonconformity score $s(x, y)$ that is small when y is a plausible label for x .
 203 This paper uses the least-ambiguous-set (LAC) score,

$$s(x, y) = 1 - \hat{p}(y | x), \quad (4)$$

204 calibrated on a held-out split $\mathcal{D}_{\text{calib}}$ of size n disjoint from both the training and test data. The
 205 calibration threshold uses the standard finite-sample correction,

$$\hat{q} = \text{Quantile}\left(\{s(x_i, y_i)\}_{i=1}^n; \frac{\lceil (n+1)(1-\alpha) \rceil}{n}\right), \quad (5)$$

206 and the prediction set for a new input x is

$$C(x) = \{y : \hat{p}(y | x) \geq 1 - \hat{q}\}. \quad (6)$$

207 Under exchangeability of $\mathcal{D}_{\text{calib}}$ and the test point, this construction guarantees

$$\mathbb{P}[y_{\text{test}} \in C(x_{\text{test}})] \geq 1 - \alpha, \quad (7)$$

208 with $\alpha=0.10$ (90% target coverage) throughout. As the scope note above states, y_{test} here is the weak
 209 label y^{KA} : Equation (7) is a real, distribution-free guarantee about agreement with the extraction
 210 pipeline, not about real-world correctness. For `dominant_tactic` and `platform`, Equations (4)
 211 to (6) apply directly over the task’s full class vocabulary. `family` and `tool` have no single multiclass
 212 structure to conformalize – each is an independent per-class binary classifier (Section 5) – so each
 213 class k gets its own binary LAC calibration and its own $\hat{q}_k$, and a sample’s per-label prediction sets
 214 are reported both individually and jointly (all labels simultaneously covered).

215 Algorithm 2 is run for 5 repeated random 60/20/20 train/calibration/test splits per task, results
 216 averaged across repeats rather than reported from one arbitrary split – a single 20% calibration set at
 217 this sample size ($n \approx 200$) is not a reliable enough estimate of $\hat{q}$ to report alone.

218 7.2 Results

219 Figure 2 plots Table 5 directly. Reading the results against the two questions this section set out to
 220 answer: first, does the guarantee hold empirically – `dominant_tactic` and `platform`’s marginal
 221 coverage (0.877, 0.925) both sit close to the 0.90 target, with set sizes well under one class on
 222 average (0.94/11, 1.19/4), i.e. the conformal sets are usually a single class or empty, not the whole
 223 vocabulary hedged defensively. Second, the conditional-coverage stress test: neither the $\text{Year} \geq 2026$
 224 nor the Conference-channel slice shows the kind of collapse that would flag a real distribution-shift

Algorithm 2 Split conformal calibration and coverage evaluation**Require:** dataset $\mathcal{D}$, task, target coverage $1 - \alpha$, repeats R

```

1: for  $r = 1, \dots, R$  do
2:   split  $\mathcal{D}$  into  $\mathcal{D}_{\text{train}}$  (60%),  $\mathcal{D}_{\text{calib}}$  (20%),  $\mathcal{D}_{\text{test}}$  (20%), stratified
3:   fit model on  $\mathcal{D}_{\text{train}}$ 
4:   compute  $\hat{q}$  on  $\mathcal{D}_{\text{calib}}$  via Equations (4) and (5)
5:   for  $x \in \mathcal{D}_{\text{test}}$  do
6:     build  $C(x)$  via Equation (6); record  $\mathbb{I}[y^{\text{KA}}(x) \in C(x)]$  and  $|C(x)|$ 
7:   end for
8: end for
9: pool test-set coverage indicators across all  $R$  repeats
10: report marginal coverage, and coverage restricted to the  $\text{Year} \geq 2026$  and Conference-channel
    slices

```

Table 5: Split-conformal coverage at 90% target ($\alpha=0.10$), pooled over 5 repeats, $n \approx 1035$ pooled test predictions per task. family/tool report mean per-label coverage and the stricter all-labels-jointly-covered rate. Set size is out of the task’s class count (K) for dominant_tactic/platform, or out of 2 (per label) for family/tool.

Task	K	Coverage (marginal)	Avg. set size	Slice: Yr ≥ 26	Slice: Conf.
Dominant tactic	11	0.877	0.94/11	0.890	0.870
Platform	4	0.925	1.19/4	0.932	0.899
Family (per-label)	18	0.912 (joint: 0.446)	0.93/2	0.487 ^{joint}	0.572 ^{joint}
Tool (per-label)	7	0.909 (joint: 0.650)	0.92/2	0.640 ^{joint}	0.752 ^{joint}

225 problem for these two tasks (all four slice numbers stay within about 3 points of the marginal rate).
 226 family and tool’s per-label coverage (0.91, 0.91) is equally close to target, but their *joint* (all-
 227 labels-simultaneously) coverage is far lower (0.45, 0.65) – an expected consequence of compounding
 228 per-label error rates across 18 and 7 largely-independent binary calibrations respectively, not a
 229 failure of Equation (7) (which was never a joint guarantee to begin with) and not itself evidence
 230 of conditional-coverage collapse, since the joint-coverage slices track the joint-coverage marginal
 231 about as closely as the multiclass tasks’ slices track theirs. Per Section 7.1, all of the above describes
 232 agreement with y^{KA} , and Section 8 below is the complementary analysis that does not carry that
 233 caveat.

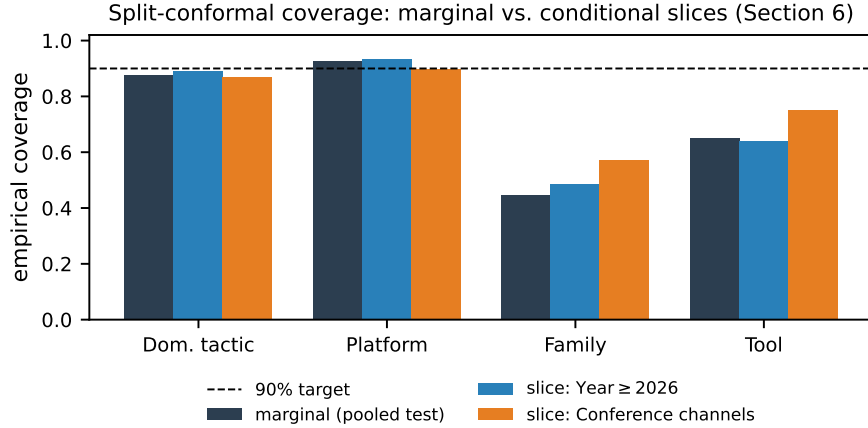


Figure 2: Empirical conformal coverage, marginal vs. the two conditional-coverage stress-test slices, against the 90% target (Table 5).

8 Out-of-Distribution Detection

Unlike Sections 5–7, this section’s results are not subject to the weak-label scope note in Section 4.3: OOD detection asks whether a model’s own confidence or embedding-space behavior differs between in-distribution and out-of-distribution input, which is answerable without knowing whether either input’s *label* is correct.

8.1 Setups and methods

Four setups, each with a known-by-construction in-distribution (ID)/out-of-distribution (OOD) split, drawn from real structure in the corpus rather than synthetic corruption:

1. **Temporal emergence.** ID: pre-2024 videos. OOD: post-2024 videos mentioning a family whose first appearance in this corpus is 2024 or later – verified directly against per-family first-occurrence year rather than assumed (Maze, Medusa, Black Basta, INC: 2024; Cactus: 2025), at Algorithm 3’s assertion step.
2. **Channel-type shift.** ID: Vendor + DFIR channels. OOD: Conference + Independent channels.
3. **Leave-rare-family-out.** ID: videos never mentioning “Cl0p” (11 occurrences total – long-tailed, unlike Conti/LockBit/ Qilin/Akira, each with 19+). OOD: the 11 Cl0p-mentioning videos.
4. **External corpus.** ID: RansKnow transcripts. OOD: 20 Newsgroups (rec.sport.baseball + sci.space), included as a sanity check that the methods below can detect an obviously unrelated domain before trusting them on setups 1–3.

Four detection methods, computed from a shared `dominant_tactic` classifier (structured features, LightGBM – Table 3’s strongest non-circular result, and a natural multiclass fit for the softmax-family methods below), refit fresh on each setup’s ID-only training pool so that no OOD signal leaks into fitting:

$$s_{\text{MSP}}(x) = 1 - \max_c \hat{p}(c \mid x) \quad (8)$$

$$s_{\text{energy}}(x) = -\log \sum_c \exp(f_c(x)) \quad (9)$$

$$s_{\text{maha}}(x) = \min_c (\phi(x) - \mu_c)^\top \Sigma^{-1} (\phi(x) - \mu_c) \quad (10)$$

$$s_{\text{conf}}(x) = 1 - \hat{p}(\hat{c}(x) \mid x) \quad (\text{Equation (4) at the predicted class}) \quad (11)$$

Equation (8) is Hendrycks and Gimpel’s max-softmax-probability baseline [Hendrycks and Gimpel, 2017]. Equation (9) is the energy score [Liu et al., 2020], computed from LightGBM’s raw pre-softmax margins $f_c(x)$ (not from $\hat{p}(c \mid x)$: probabilities already sum to 1, which makes Equation (9) degenerate to a constant if computed from them directly – a real pitfall caught while implementing this section, not a hypothetical one). Equation (10) is the class-conditional Gaussian Mahalanobis distance of Lee et al. [2018], with $\phi(x)$ the 384-dim mean-pooled sentence embedding (Section 5’s footnote) and μ_c, Σ the ID-train class-conditional means and pooled covariance, using the classifier’s own *predicted* classes to form groups (consistent with Section 4.3: no independently verified labels exist to group by instead). Equation (11) reuses Section 7’s LAC score, evaluated at the predicted rather than true class for OOD flagging – this is deliberately, not accidentally, the same quantity as Equation (8); the two are reported separately for comparability with Equation (5)’s calibrated threshold (does a point’s own score exceed the 90%-target $\hat{q}$), which is the operationally distinct number.

The assertion in Algorithm 3 is not decorative: the temporal- emergence setup’s family list was originally drafted from memory as “Qilin, DragonForce, Black Basta,” before checking actual first-occurrence years against the corpus. Qilin’s first real occurrence is 2019, not post-2023, and DragonForce does not appear in this corpus at all – both would have silently produced a meaningless setup (there is no real ID/OOD separation if the “emerging” families already appear throughout the ID training period). The corrected list (Maze, Medusa, Black Basta, INC, Cactus) is the one actually used in Table 6 and is verified by Algorithm 3 on every run, not just checked once by hand.

Algorithm 3 OOD setup evaluation

Require: corpus $\mathcal{D}$, ID/OOD partition rule for the setup

- 1: **assert** every claimed-emerging family’s first-occurrence year matches the setup’s stated cutoff $\triangleright$ caught a factual error at this step – see note below
 - 2: split ID pool into $\mathcal{D}_{\text{train}}$ (75%) / $\mathcal{D}_{\text{test}}^{\text{ID}}$ (25%)
 - 3: fit `dominant_tactic/structured/LightGBM` on $\mathcal{D}_{\text{train}}$
 - 4: score $\mathcal{D}_{\text{test}}^{\text{ID}}$ and $\mathcal{D}_{\text{OOD}}$ under Equations (8) to (11)
 - 5: report AUROC(ID vs. OOD) per method, plus the conformal exceedance-rate contrast
-

8.2 Results

Table 6: OOD-detection AUROC by setup and method (Equations (8) to (11)). $n_{\text{ID}}/n_{\text{OOD}}$ are ID-test/OOD sample counts. Mahalanobis is unavailable for `external_corpus` (20 Newsgroups has no entry in the RansKnow-only embedding cache).

Setup	n_{ID}	n_{OOD}	MSP	Energy	Mahalanobis	Conformal
Temporal emergence	55	31	0.561	0.360	0.223	0.533
Channel-type shift	156	353	0.477	0.487	0.666	0.487
Leave-rare-family-out	256	11	0.377	0.350	0.145	0.376
External corpus (sanity)	259	1190	0.625	0.476	–	0.631

Figure 3 plots Table 6. The honest reading is a mixed and mostly negative one, which is itself the finding worth reporting rather than a result to explain away. Three observations:

The confidence-based methods (MSP, energy, conformal) hover near chance (0.5) on every setup, including the sanity check. Diagnosing *why* on `external_corpus` is informative: 99.6% of 20 Newsgroups documents are classified None (no dominant tactic detected) by the shared classifier, and with *higher* average confidence than the real in-distribution test set (mean $\hat{p}_{\text{max}}$: 0.921 OOD vs. 0.895 ID). A near-empty, off-topic TF-IDF vector and a genuinely tactic-free ransomware transcript both produce the same signature – confidently “no tactics detected” – so a classifier whose decision function is fundamentally an argmax over small keyword-mention counts has no way to distinguish “nothing here” from “wrong domain entirely.” This is a known blind spot of softmax/probability-based OOD detection in general [Hendrycks and Gimpel, 2017], reproduced concretely here rather than assumed from the literature.

Mahalanobis distance on real sentence embeddings is the one method that separates classes at all, and only partially. It reaches 0.666 on channel-type shift – Conference/Independent channels differ enough in embedding space (production style, framing, audience) to be mildly detectable – but performs *worse* than chance on temporal emergence (0.223) and leave-rare-family-out (0.145). Both of those setups’ OOD population describes the same underlying phenomenon (ransomware incidents, attacker tradecraft) as the ID population, just naming a different or newer group – exactly the case where sentence embeddings, which encode semantic content rather than proper nouns, would *not* be expected to separate the two populations well. The small OOD sample sizes in these two setups ($n=31$, $n=11$) also mean each AUROC estimate carries substantial variance on its own; with $n_{\text{OOD}}=11$, a single re-ordered pair moves the estimate by roughly 1/11.

The conformal exceedance-rate contrast (Equation (5)’s $\hat{q}$) is uninformative in this corpus specifically. In every setup, the calibrated threshold $\hat{q}$ lands so close to the maximum possible score ($\hat{q} \geq 0.99$ in three of four setups) that essentially no point, ID or OOD, exceeds it – a consequence of the classifier being very confidently correct on the bulk of its small ($n \approx 115$) calibration split, which pushes the 90th-percentile score to an extreme value. Equation (11)’s AUROC column is the more informative number from this method here; the exceedance-rate contrast is reported in the released JSON (outputs/phase4_ood.json) for completeness but is not a useful discriminator at this sample size.

Taken together, Section 8.2 does not support a general claim that this pipeline’s models reliably detect distributional shift; it supports a narrower, still useful one: OOD detection here is hard, softmax-confidence-based methods have an identifiable and reproducible blind spot on this specific classifier (the “no signal” vs. “wrong domain” confusion above), and embedding-distance methods help only

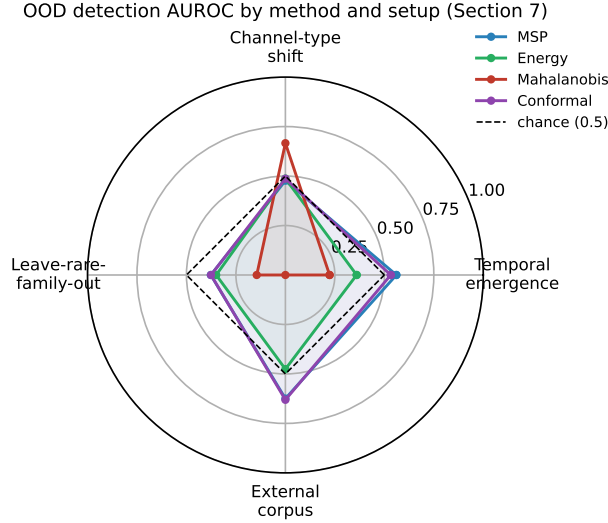


Figure 3: OOD-detection AUROC by method (color) and setup (axis), against the chance line at 0.5 (Table 6). Mahalanobis is the only method to clear chance by a wide margin, and only on one of four setups.

when the shift is stylistic rather than topical. Unlike Sections 5 to 7, none of this depends on whether y^{KA} is correct (Section 4.3) – it is a direct, if modest, empirical finding about model behavior.

9 Interpretability and Perturbation Robustness

Unlike Sections 5–7, this section asks questions about model *behavior* – does it exploit shortcuts, what does it rely on, how much do its predictions move under semantically-irrelevant edits – none of which require independent ground truth to answer, so Section 4.3’s scope note does not apply here.

9.1 Confound checks

Length. Transcript word count correlates moderately with `Tactic_Total_Mentions` (Pearson $r = 0.409$) and weakly with `Family_Count` ($r = 0.131$), but not at all with `Tool_Total_Mentions` ($r = 0.043$, not significant) – the confound is not uniform across signal types. Because it is mathematically guaranteed that dividing every tactic’s count for one video by that video’s own word count cannot change which tactic has the highest count (argmax is scale-invariant per row), the confound cannot affect how the `Dominant_Tactic` label itself was derived; the question that matters is whether it inflates `dominant_tactic`’s strongest result (structured features, Table 3). Re-scoring with length-normalized (mentions per 1,000 words) instead of raw-count features gives 0.753 macro-F1 against the raw-count model’s 0.764 – a 1.1-point difference, not a length-exploitation artifact.

Channel identity. 486 of 1,034 transcripts (47%) contain at least one literal mention of their own channel’s name. Masking those mentions and re-scoring TF-IDF LogReg under both CV protocols leaves the `channel_grouped-vs-stratified_random` gap unchanged for `family` (0.028 raw $\rightarrow$ 0.031 masked) and for `dominant_tactic` (0.020 raw $\rightarrow$ 0.029 masked). Whatever a model exploits when a channel is not held out, it is not the channel literally naming itself – the leakage this split protocol exists to catch is stylistic or topical, not addressable by name-masking.

9.2 Counterfactual explanations

Motivated directly by Finding A: does a classifier trained on the *fixed* labels still show the regex’s original word-collision fragility? A TF-IDF+LogReg model was trained on today’s corrected `Family` labels and evaluated on the 275 videos Section 4.1 confirmed as false positives under the pre-fix regex

341 (videos where “play” appears without ransomware context). The trained model assigns essentially
342 zero probability to Play on all 275 (maximum $P = 0.013$) – it did not inherit the bare-word-collision
343 bug.

344 The complementary question – what drives the model’s actual positive Play predictions – was then
345 asked with a greedy word-*type* removal counterfactual search (remove the vocabulary word whose
346 removal most reduces $P(\text{Play})$, repeat until the prediction flips or a budget is exhausted) on the 8
347 highest-confidence true positives. play appears in 5 of 8 removal sets and is consistently one of
348 the highest-impact single removals when present, but **0 of 8 cases flip on removing “play” alone** –
349 every flip required 4–10 additional removals, most of them generic (function words, not ransomware-
350 specific vocabulary). The model reduced, but did not eliminate, reliance on the single ambiguous
351 trigger word, and the words it combines that reliance with are not obviously a clean, interpretable
352 “ransomware vocabulary” set – worth flagging as its own finding about decision-boundary diffuseness,
353 independent of the word-collision question it was designed to answer.

354 9.3 Perturbation robustness

355 Untargeted counterpart to the counterfactual search above: two perturbation types, measuring flip rate
356 (does the top prediction change) and mean absolute probability drift on freshly-fit TF-IDF+LogReg
357 models. *Simulated ASR noise* (compound proper nouns split apart, e.g. WannaCry→“wanna cry,”
358 Mimikatz→“mimi cats,” plus 8% random token deletion) was applied to the 74 official-caption
359 videos specifically – the “clean” subset, simulating what those transcripts would look like if they had
360 gone through the ASR cascade that produced the other 92.7% of the corpus (Finding B). *Synonym*
361 *substitution* (common non-jargon words swapped for synonyms) was applied to a 150-video sample.

362 dominant_tactic, platform, and relevance are essentially unmoved by either perturbation
363 (0.0–0.7% flip rate, mean drift < 0.01 in every case). family flips 20–26% of the time under
364 both perturbations despite a much smaller mean drift (~ 0.002) than any other task – not a new
365 problem, but the same weak-model issue from Table 3 (family’s best macro-F1 is 0.129) surfacing
366 differently: closely-bunched, low-confidence probabilities mean a negligible nudge is enough to
367 change which barely-ahead class wins. On the framing of “adversarial” robustness for this dataset:
368 no real adversary submits transcripts to evade this classifier in production, so a full gradient-based
369 adversarial-attack framework would be disproportionate to what this paper needs; the actual risk
370 surface is ASR transcription noise and channel-style variation, which is what this sweep directly
371 measures. Combined with Section 8’s temporal-emergence and leave-rare-family-out setups, these
372 results are the paper’s answer to how the system handles evolving, novel, and adversarial-adjacent
373 input – not three separate open questions.

374 10 Limitations and Ethics

375 **[TODO]** Expand into full prose; bullets below are the content, not the final form. The first bullet is
376 the paper’s central limitation and should probably open this section as prose, not sit inside a list with
377 four other items of noticeably smaller consequence.

- 378 • **No independent ground truth.** This is the paper’s central limitation, not one bullet among
379 several, and is why it is stated repeatedly rather than once: this project does not include a
380 human-annotated evaluation set independent of the Knowledge Agent pipeline (Section 4.3).
381 External annotator recruitment required institutional ethics approval not available to this
382 project; a single-investigator annotation pass was considered and rejected as still too close to
383 the system under evaluation to count as independent evidence. Every result in Sections 5–7
384 is consequently agreement with an audited but demonstrably imperfect extraction pipeline
385 (Finding A), not verified accuracy – Section 8’s OOD results and Section 9’s behavioral
386 diagnostics are the exceptions, since neither depends on label correctness. Recruiting
387 independent annotators, whether through formal ethics approval or an unpaid collaborator
388 arrangement that falls outside whatever definition of human-subjects research applies at
389 the author’s institution, is the single highest-value extension of this work – more valuable
390 than any additional modeling in Sections 5–7 would be, since none of that modeling can
391 currently be checked against anything but itself.

- **Sampling bias.** The corpus is conditioned on which channels operators chose to run, which videos they chose to title and describe in keyword-matchable ways, and YouTube’s own recommendation and search surfacing – not a random or exhaustive sample of ransomware-relevant discourse.
 - **ASR noise.** 92.7% of transcripts are machine-transcribed, mostly by `whisper_base`. Section 9.3’s simulated-ASR-noise sweep bounds how much this moves predictions; Finding B’s word-error-rate-on-jargon question (comparing ASR output against source audio directly, rather than a simulation of ASR-style errors) remains open.
 - **Weak-label ceiling.** Every quantitative result in Sections 5–7 is bounded by the Knowledge Agent’s own precision/recall, which Finding A shows is not perfect even after the disambiguation fix, and which Section 9.2 shows partially propagated into at least one trained model.
 - **Dual use.** This is a corpus of ransomware tactics, tools, and case studies extracted from public educational content.
- [TODO]** Add a considered dual-use statement – the underlying source material is already public (YouTube videos aimed at defenders), but structuring it may lower the effort to extract offense-relevant patterns; discuss how this compares to existing public MITRE ATT&CK mappings and threat-intel reporting, which already aggregate similar information.

11 Conclusion

[TODO] Write last. Should state the label-quality audit methodology and code (Section 4, Section 9) as a second contribution alongside the transcripts themselves – other weak-label security-NLP datasets built the same way (keyword/regex extraction over scraped text) can reuse the audit methodology even without reusing the corpus. Should also state candidly that the most valuable open follow-up to this specific paper is independent human-annotated verification (Section 10), not further modeling – and should say why explicitly rather than trailing off: every result this paper reports beyond Section 8 would change in *meaning*, not just in number, the day that verification exists.

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437 **A Keyword List**

438 **[TODO]** Insert the full KEYWORDS list from `Scripts/fetch_transcripts.py` verbatim.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract and introduction (once finalized – see remaining \todo markers) state the corpus, the label-quality audit methodology and findings, weak-label baseline results explicitly framed as pipeline-agreement rather than verified accuracy, and OOD-robustness results as the paper’s contributions, matching Sections 3–8.

Guidelines:

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Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

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Justification: Section 3 specifies the channel/video selection criteria and full keyword list (Appendix A); Section 5 specifies the task taxonomy, feature representations, models, and CV protocols. All extraction, audit, and experiment code is in the public repository (see next item), including the exact hyperparameters used (default scikit-learn/LightGBM settings unless otherwise noted in code).

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551 Question: Does the paper provide open access to the data and code, with sufficient instruc-
552 tions to faithfully reproduce the main experimental results, as described in supplemental
553 material?

554 Answer: [Yes]

555 Justification: Dataset: [https://www.kaggle.com/datasets/henrykabuye/](https://www.kaggle.com/datasets/henrykabuye/ransknow-v1)
556 [ransknow-v1](https://www.kaggle.com/datasets/henrykabuye/ransknow-v1) (CC0-1.0). Code: <https://github.com/Henrinnes/RansKnow>
557 (fetch pipeline, label-quality audit scripts, rk_pipeline experiment framework, Phase 5
558 confound/counterfactual/perturbation scripts), released under the MIT License (LICENSE in
559 the repository root).

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582 rameters, how they were chosen, type of optimizer) necessary to understand the results?

583 Answer: [Yes]

584 Justification: Section 5 specifies the three CV protocols (stratified random, channel-grouped,
585 temporal) and the five-model/three-representation grid; exact hyperparameters (regulariza-
586 tion, tree counts, TF-IDF vocabulary size, embedding model and chunking) are documented
587 in Scripts/rk_pipeline/ and referenced in Section 5.

588 Guidelines:

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591 that is necessary to appreciate the results and make sense of them.
- 592 • The full details can be provided either with the code, in appendix, or as supplemental
593 material.

594 7. Experiment statistical significance

595 Question: Does the paper report error bars suitably and correctly defined or other appropriate
596 information about the statistical significance of the experiments?

Answer: [No]

Justification: Reported macro-F1 values are single-seed, 5-fold (or single-split, for the temporal protocol) point estimates, not averaged or bounded across repeated random seeds.

[TODO] This is a real, addressable gap, not an inherent limitation – the framework’s split protocols already accept a seed parameter; re-running the Phase 1 grid across 3–5 seeds and reporting standard deviation is worth doing before submission, given how cheap most combinations were to run (see compute-resources item below).

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8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: Experiments ran on a local workstation: 32-core CPU (structured/TF-IDF representations, all sklearn/LightGBM models) plus 3×NVIDIA GTX 1070 Ti, 8GB each (embedding representation, sentence-transformer encoding). Observed wall-clock time: individual task/model/split combinations ranged from seconds to ~80s for the heaviest (RandomForest on 18-class multilabel TF-IDF); full grids (5 tasks × up to 5 models × 3 splits) completed in 20–45 minutes per representation.

[TODO] State total compute across the whole project, including exploratory runs that didn’t make the paper, per the guideline – not tracked precisely during development.

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Justification: The authors have reviewed the NeurIPS Code of Ethics. The corpus is built from already-public YouTube content (Section 3); the current scope includes no human-subjects data collection (Section 4.3); dual-use considerations are discussed in Section 10.

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10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: Section 10's dual-use bullet discusses both the corpus's positive use (ransomware defense research, DFIR education) and the risk that structuring defender-oriented content could lower the effort to extract offense-relevant patterns.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
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